

Ryan Camilleri

Observational cosmologist

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Summary

I am a postdoctoral research fellow at The University of Queensland specialising in observational cosmology, with a focus on standard candle techniques using Type Ia supernovae and tests of beyond- Λ CDM models. I have strong experience building end-to-end cosmology pipelines for data analysis, simulation, and parameter inference alongside statistical analysis and model testing. I am also proficient working with established cosmological software frameworks such as COSMOSIS, SNANA, and COBAYA.

Education

Doc- tor of Phi- loso- phy	The University of Queensland , Supernova Cosmology with the Dark Energy Survey	Brisbane, Australia June 2022 – Sept 2026
Bach- elor of Sci- ence (Hon- ours)	The University of Queensland , Physics	Brisbane, Australia Feb 2021 – June 2022
Bach- elor of Sci- ence	Griffith University , Physics & Mathematics	Gold Coast, Australia Feb 2017 – Dec 2020

Awards

Honourable Mention for Best Student Publication in Physics Awarded at The University of Queensland	2025
Dark Energy Survey “Builder” Recognition for contributions to the DES Collaboration	2025
Best Presentation - non-invited Speaker Awarded at the Frontiers and Gravitational physics conference in Portsmouth, England.	2024
Head of School Award for Excellence in Science Awarded to the Griffith University student in their final year with the highest career cumulative GPA in the Bachelor of Science (GPA 7.0).	2020
Australian Institute of Physics Prize Awarded to the most outstanding graduating student in a Physics major (GPA 7.0).	2020
Griffith Futures Scholarship Awarded to students who have been impacted by various types of educational disadvantage.	2020

Griffith Award for Academic Excellence	2017
Awarded to students who achieve a cumulative GPA of at least 6.0, calculated over the academic year. I received this award each year of my undergraduate degree.	

Publications

Supernova Unite: Constraints on cosmologies beyond-ΛCDM	2026
In preparation R. Camilleri et al.	
Supernova Unite: Combining Pantheon+ and DES-SN5YR	2026
R. Camilleri et al. (Submitted to Publications of the Astronomical Society of Australia)	
Supernova Unite: Host-Galaxy Mass Measurements of Type Ia Supernovae and Their Impact on Cosmology	2026
J. lee, R. Camilleri et al. (Submitted to Astrophysical Journal)	
DES Supernova Program: H_0 from the Inverse Distance Ladder	2025
R. Camilleri et al. (Monthly Notices of the Royal Astronomical Society)	
DES Supernova Program: Investigating Beyond-ΛCDM	2024
R. Camilleri et al. (Monthly Notices of the Royal Astronomical Society)	
DES Supernova Program: Reanalysis of Cosmology Results	2025
B. Popovic et al. (Monthly Notices of the Royal Astronomical Society)	
Dark Energy Survey: implications for cosmological expansion models from the final DES Baryon Acoustic Oscillation and Supernova data	2025
DES Collaboration (Physical Review D)	
Cosmological non-linearities as an effective fluid	2025
L. Giani, R. v. Marttens, R. Camilleri (Physical Review Letters)	
WiFeS observations of nearby southern Type Ia supernova host galaxies	2024
A. Carr, T. M. Davis, R. Camilleri et al. (Publications of the Astronomical Society of Australia)	

Activities

UQ-QASMT Research Mentoring Program	July 2025 – 2026
I guided a group of Year 10 students through an independent cosmology project, introducing them to Python coding, and helping them develop the skills to produce clear plots and reports that effectively communicate their results.	
Dark Energy Survey	2022 – 2025
I organised and led tri-weekly international meetings, facilitating collaboration between Australia and UK teams.	

UQ Astro Work Experience Program

2022 – present

The Astronomy group at The University of Queensland runs a weeklong work experience program (typically mid-year) and offers future STEM students who want careers in astronomy a first-hand look at the day-to-day life of an Astronomer. The week consists of an astronomy project, telescope demonstrations, “Meet the researcher” discussions and on the final day, the students get to present what they have learned to the group. Since 2022, I have volunteered to help organise and run this program.

Conferences

Next Generation of SN Surveys Mini-Workshop

Presented my paper: Supernova Unite: Combining Pantheon+ and DES-SN5YR

Oxford, United Kingdom

July 2026

Supernova Cosmology: New Challenges and Opportunities

Presented my paper: Supernova Unite: Combining Pantheon+ and DES-SN5YR

Seoul, South Korea

June 2026

The End of Λ ?

Invited talk presenting both the DES supernova cosmology results and my current project, which is a combined DES & Pantheon+ analysis

Incheon, South Korea

Sept 2025

Cosmic Flows

Presented H_0 inverse distance ladder results

Brisbane, Australia

Feb 2025

DES Spring Meeting

Presented beyond- Λ CDM paper

Barcelona, Spain

May 2024

Frontiers in Cosmology & Gravitational Physics

Presented beyond- Λ CDM paper

Portsmouth, England

May 2024

OzGrav ECR Winter School

Attended the OzGrav ECR Winter School

Kingscliff, Australia

Aug 2025

Memberships

Dark Energy Survey (DES) Collaboration: Builder. The Dark Energy Survey is an international, collaborative effort to map hundreds of millions of galaxies, detect thousands of supernovae, and find patterns of cosmic structure that will reveal the nature of the mysterious dark energy that is accelerating the expansion of our Universe.

ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav): Member. The mission of OzGrav is to capitalise on the historic first detections of gravitational waves to understand the extreme physics of black holes and warped spacetime, and to inspire the next generation of Australian scientists and engineers through this new window on the Universe.

Dark Energy Spectroscopic Instrument (DESI) Collaboration: Member. DESI seeks to map the large-scale structure of the Universe over an enormous volume. Targeting about 30 million pre-selected galaxies across one-third of the night sky, scientists will use DESI’s redshifts data to construct 3D maps of the universe.

References

Prof. Tamara Davis

PhD Supervisor, University of Queensland — tamarad@physics.uq.edu.au

A/Prof. Maria Vincenzi

DES Supernova Working Group Convenor — maria.vincenzi@physics.ox.ac.uk

Dr Paul Shah

DES Collaboration — paul.shah.19@ucl.ac.uk